

Leah Operana

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SUMMARY

Entry-level Software Engineer with additional experience as a Technical Program Lead with a successful track record of delivering large-scale STEM initiatives impacting 25,000+ users annually through end-to-end program management, curriculum design, and quality assurance. Experienced in software development (Python, Java), API integration, distributed systems, cross-functional collaboration, and machine learning.

PROJECTS

mini-SWE-Agent, Python, Bash, Docker, Miniconda, GitHub, Gemini 2.5 Flash API, SWE-Bench

- Developed LLM coding agent framework that autonomously reads, writes, and modifies code to solve real-world software-engineering tasks, leveraging Gemini LLM API to generate code patches for real GitHub issues
- Achieved a performance score of 127% on the SWE-bench Verified-40 subset by resolving 28 out of 40 issues, exceeding the required 22 issues; collaborated effectively in a team, enhancing project management and communication skills

AI-Powered Weather Assistant – Product Manager, Java, Gemini AI, REST API

- Led cross-functional team of 6 developers full product lifecycle, defining product roadmap and prioritizing features including Gemini AI integration and real-time weather alerts to deliver MVP within semester timeline
- Drove technical requirements gathering and API selection process, evaluating OpenWeatherMap vs. competitors and defining integration strategy for Google Gemini AI to enable conversational weather queries
- Facilitated Agile ceremonies

Ant Colony Simulation, Java

- Engineered a multi-agent emergent behavior simulation using recursive algorithms and OOP, modeling realistic ant colony dynamics across configurable environment parameters
- Optimized memory allocation and state update logic to improve simulation performance by 35%, earning a 100% project score

Theo's World (2D Adventure Game), Python, Pygame

- Designed a 2D game engine with an event-driven input system, frame-rate-controlled rendering pipeline, and rectangle-based collision detection to support responsive real-time gameplay
- Managed asset loading, surface abstraction, and sprite positioning logic to produce a functional animated game loop within a modular codebase

EXPERIENCE

TATA CONSULTANCY SERVICES (TCS), Chicago, IL

Aug 2022 – Present

One of the world's largest IT services and consulting firms, TCS specializes in AI-led digital transformation, cloud computing, and cybersecurity across 50 industries globally, generating \$30B+ in annual revenue.

North America STEM Program/Data Lead

Reporting to the Global Head and North America CSR Head, responsible for managing a team of 1-3 direct reports and overseeing the full lifecycle of TCS's North America STEM programming and data operations. Accountable for partner organization outcomes, data and analytics deliverables, and large-scale event execution across the region.

- Scaled data infrastructure supporting 20,000+ users annually across free STEM programs for underserved youth, redesigning data models to improve platform reliability
- Built automated Excel dashboards to centralize program tracking and reporting across the team, cutting manual data management time by 50%+ and reallocating bandwidth toward direct community partner engagement
- Translated partner and stakeholder feature requests and event logistics into technical specifications – using flowcharts and requirement summaries to align external vendors and program/ops teams – reducing deployment timelines by 25% across North America STEM initiatives
- Planned and executed 20+ large-scale tech events in partnership with schools and nonprofits serving underrepresented communities – troubleshooting live technical systems, coordinating logistics end-to-end, and delivering hands-on training in AI and software development to expand participants' technical skills

C2 EDUCATION, Naperville, IL**May 2019 – Aug 2022**

A nationally franchised K-12 education services company, C2 Education delivers personalized tutoring, test preparation, and STEM instruction to students and families across 200+ centers in the United States

Education Coordinator & STEM Instructor

Reporting to Center Director, responsible for coordinating center operations, leading tutors and instructors, and delivering structured STEM

- Engineered a QR code-based scheduling system that automated client communication by triggering pre-populated SMS messages upon scan, reducing scheduling errors by 60% and eliminating manual back-and-forth between clients and center staff
- Designed and delivered daily lesson plans integrating Python to teach coding fundamentals alongside science and test prep concepts – driving measurable outcomes including improved test scores, student level advancement, and completed coding projects – consistently earning 100% positive parent feedback across 3-year tenure

EDUCATION

Master of Computer Science • University of Illinois Urbana-Champaign (In Progress)

Bachelor of Science • Psychology and Pre-Med • Loyola University Chicago – Magna Cum Laude

SKILLS

Python • NumPy • Pandas • Java • AI/ML • OOP • SQL • Data Structures • Algorithms • LLM APIs • API Development • Debugging/Testing • Teaching • Collaboration